



CASCAD-e PROJECT STANDARDIZATION ACTIVITIES

API Design and possible impact in LF CAMARA

Turin, February 26th, 2026

CASCAD-e project

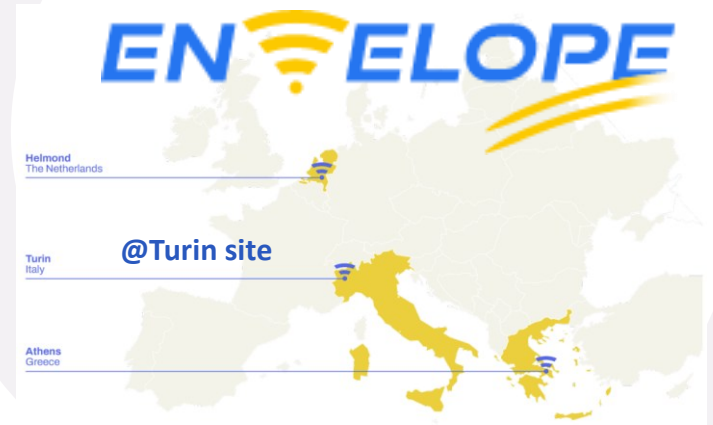
Connected Ambulance Safety with Computation and Advanced Deployment at the edge

Project timeline: Dec 2025 – May 2026

Small EU consortium (NEXT G CLOUD, Smart Flow, Wings Ict Solutions).

Tentative Advisory Board (Croce Rossa Italiana, ANPAS, AOU Cagliari, XRSI Europe, 5GAA)

<https://envelope-project.eu/>

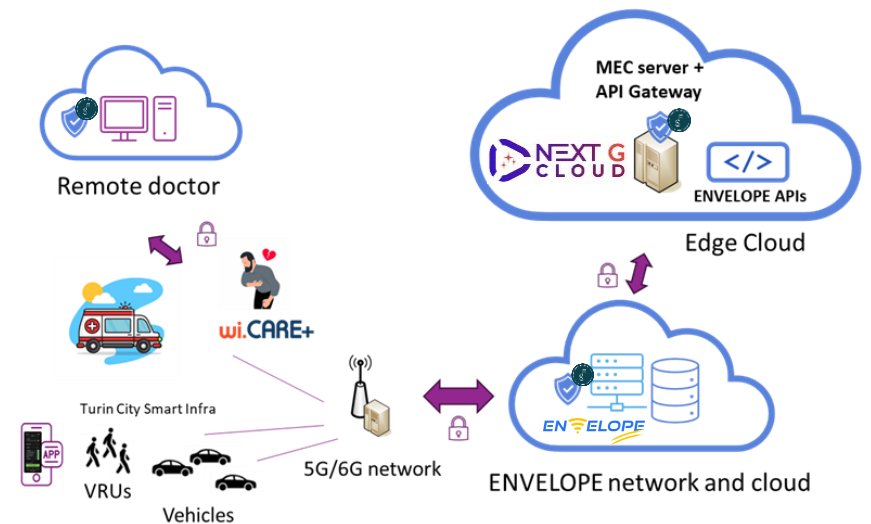


Service use cases:

- **Collision Risk Detection and Safe Routing for Emergency Vehicles** – real-time traffic data will be used to predict the drivers' behaviour, and critical alerts will be sent to nearby vehicles and VRUs, improving safety and speed of intervention in dense areas or during mass events
- **Real-Time Remote Triage and Diagnostics** – a connected ambulance equipped with high-resolution cameras, medical sensors, and edge computing capabilities can stream patient vitals, ECGs, and high-definition video to emergency physicians at the hospital in real time
- **Secure Biometric and Incident Data Upload to Electronic Health Records (EHR)** – during patient transport, biometric and contextual data (e.g., accident scene video, environmental conditions) can be processed and temporarily stored at the edge for immediate decision support. Once securely validated, relevant data is selectively uploaded via trusted APIs to the patient's EHR, using privacy-aware access controls and real-time encryption mechanisms. This ensures continuity of care while respecting privacy and regulatory compliance.

User Story:

- wi.CARE+ app: **advanced triage** during calls to the emergency service
- MEC server supported by the ENVELOPE's API collects **events** on locations of ambulances and patients
- The closest ambulance is sent to the caller's location
- Road users receive **notifications** along the ambulance route
- wi.CARE+ app is used on board of the ambulance to allow a **consultancy with a remote doctor** during the transportation
- The patient arrives to the appropriate hospital
- **Privacy and data security** is maintained and guaranteed throughout the entire procedure



GENERAL BACKGROUND

*Next G Cloud is currently involved in two different projects in Turin related to **edge computing** and **smart use cases***



The logo for the ToMove project, featuring the word "ToMove" in a stylized font. "To" is green and "Move" is dark blue.

Next G Cloud acts as the **prime contractor**, with the objective of demonstrating and developing, during the trial phases, all aspects of **smart mobility**, with particular attention to **vulnerable road users**.

Project partners:

- **QuandoPasso**, developer of the **DriveBud application**, which enables interaction and communication between **drivers, pedestrians, motorcyclists, and other road users**.
- **X Flow Research**, (as sub-contractor of Next G Cloud) is responsible for **software development**, with a primary focus on **API development**



The logo for the EN ELOPE project, featuring the text "EN ELOPE" in blue with a yellow Wi-Fi symbol between "EN" and "ELOPE".

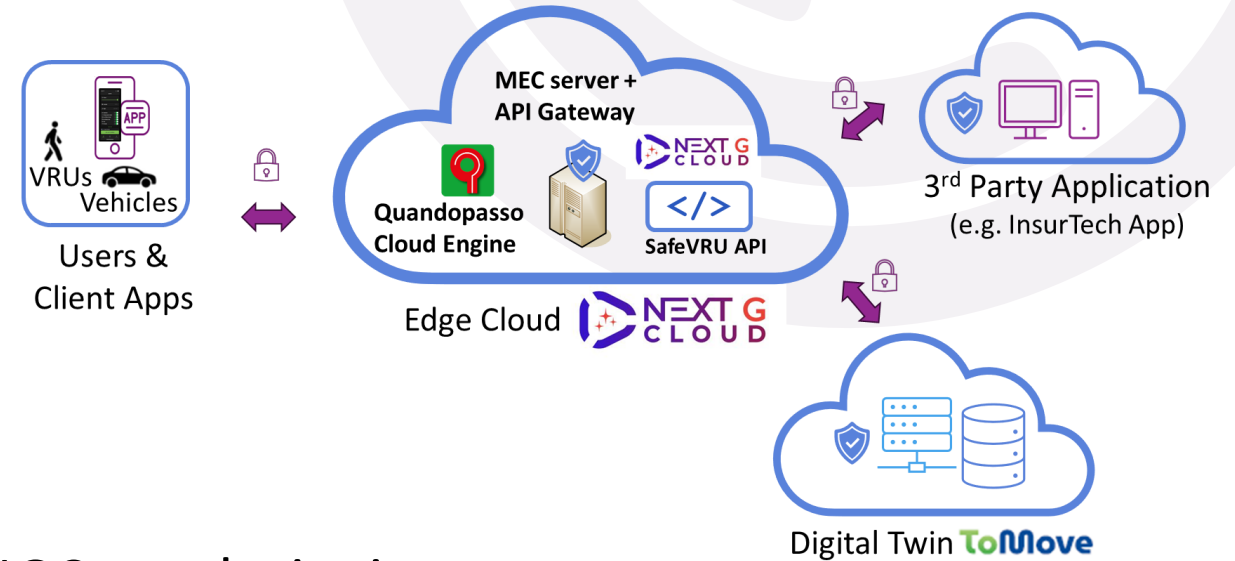
Next G Cloud acts as the **prime contractor**, demonstrating the proof of concept of a **connected ambulance** and the **Tele-medicine**, ensuring data security.

Project partners:

- **SmartFlow** focuses on **DPO agility**.
- **Wings ICT Solutions** provides the **platform and the tool** to be deployed in the ambulance and other devices.
- **X Flow Research** (as sub-contractor of Next G Cloud) is responsible for the **full API software development**.

ToMOVE: High-level system architecture

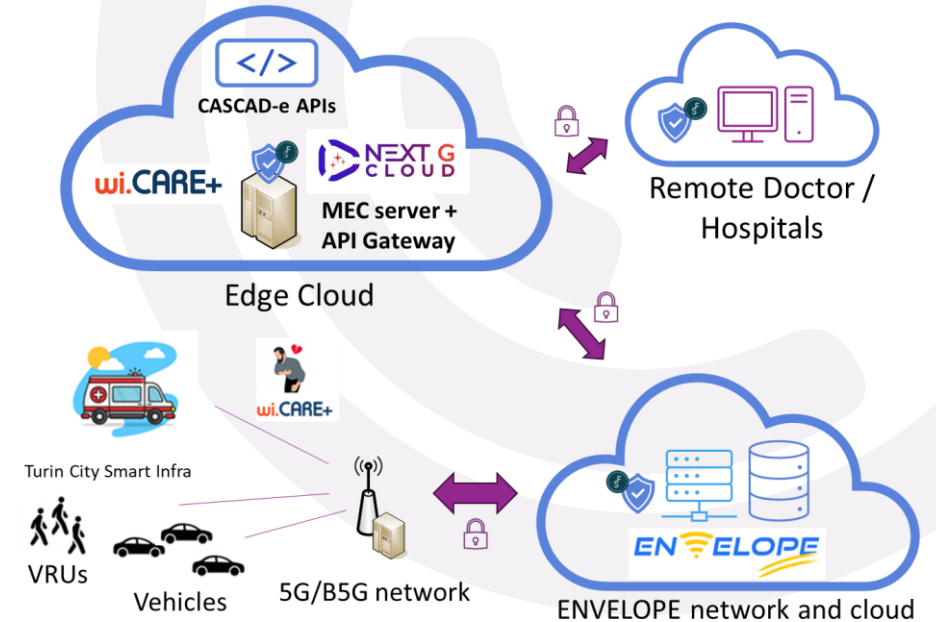
- The starting point of our infrastructure is based on the edge cloud developed for ToMOVE (with API GW and MEC)



- In ToMOVE, xFlow and NGC are designing an initial API (SafeVRU API) for the targeted *automotive* use cases

CASCAD-e SYSTEM DIAGRAM

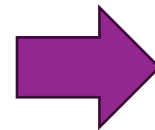
- The system developed for CASCAD-e reuses API Gateway and ETSI-compliant MEC platform (from TOMOVE) to expose enhanced functionalities at the edge



CASCAD-e APIs

- In CASCAD-e, xFlow and NGC are designing enhanced APIs (as evolution of SafeVRU API) for the targeted use cases on connected ambulance.

The opportunity for the dissemination that we see



Propose it as a new API in CAMARA project

THE ASK

- We are seeking for support on this initiative to propose a new API in CAMARA project.
 - ENVELOPE stakeholders also involved in CAMARA may support such presentation at a forthcoming CAMARA meeting
 - Turin City may approve the synergy with ToMOVE and allow NEXT G CLOUD to use the Digital Twin datasets also for the CASCAD-e project.

Advantages:

- Outstanding dissemination/Standardization for ENVELOPE consortium
- Great visibility for ToMOVE project and the Turin City
- CAMARA companies may have a better perception of an API actually designed for real use cases in a Smart City.

X FLOW – API PROPOSAL

MEC-CAMARA Collaboration for V2X Service Exposure

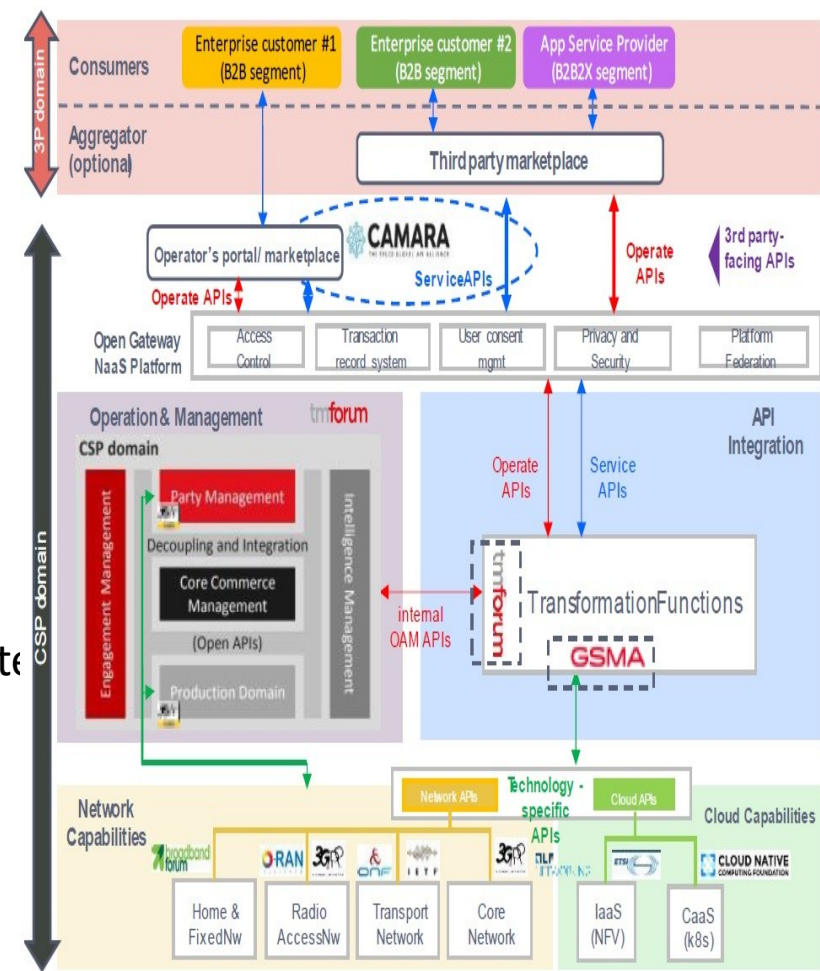
ETSI MEC and CAMARA complement each other to enable standardized and scalable V2X service exposure. It explains how MEC provides low-latency edge execution and contextual awareness, while CAMARA offers developer-friendly, common northbound APIs for accessing network and edge capabilities. The collaboration bridges edge-native services and cloud-centric API exposure, enabling interoperable V2X use cases such as cooperative awareness, hazard notifications, and real-time traffic services across operators and ecosystems.

[Presented at MEC Plenary#44](#)

MOU FRAMEWORK OF COLLABORATION

- Complementary work on northbound and southbound API exposure
 - CAMARA → focuses on NBI “Service APIs” - developer-friendly abstractions.
 - ETSI MEC → provides standardized internal APIs – southbound APIs (edge-native).
- Joint efforts include:
 - Align standards and API definitions between ETSI MEC and CAMARA.
 - API exposure and interworking guidance, with ETSI White PAPER No. 68 as a key reference.
 - Other areas of collaboration may include the work item MEC 043 on Abstracted API for Industries.

https://docbox.etsi.org/Partners/Agreements/Linux_Foundation_MoU_2022.pdf



3rd Party-facing APIs

Service APIs

App-centric, developer-oriented
Apache2.0 lic, user -friendly , easy-to-use
Example: QoD, verifylocation, device status, Sim Swap, ...
Includes some management functionality used from the apps (in-app OAM APIs)

Hosted by **CAMARA**

Contributed by OpenGateway partners , directly or supported by bodies like



Operate APIs

Management oriented
Easy-to-implement , easy-to-use, simple
Example: register, account, monitor, issue mgmt, order/purchase, pay...
Provides an easy integration of the NaaS Platform with marketplaces /portals

Contributed by OpenGateway partners, hosted by **tmforum**

Technology -specific APIs

Technical capability oriented, standard, (FRAND) deterministic
Example: policysetting parameter setting information check...

Contributed by specific domain SDOs



MOTIVATION

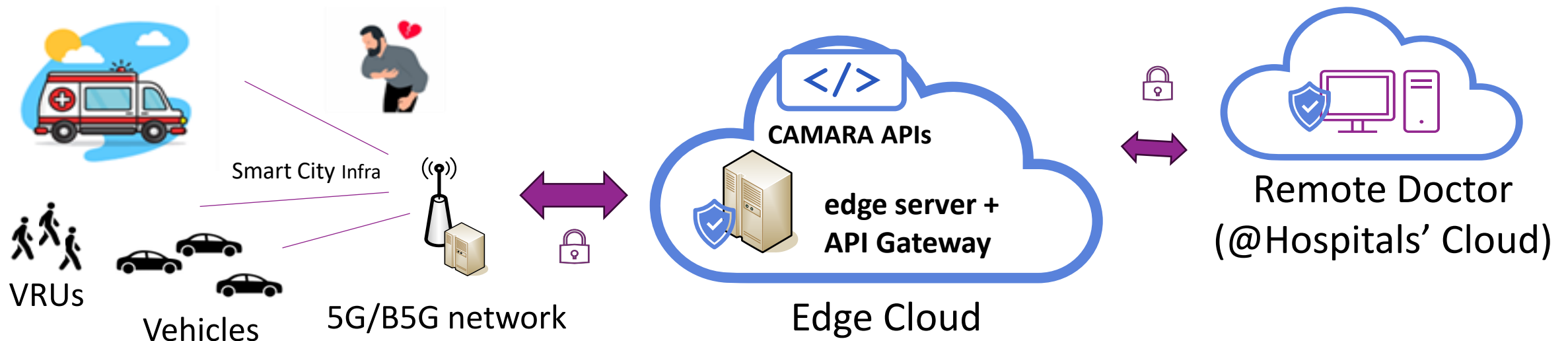
- Vehicles need real-time intelligence including QoS/QoD prediction, connectivity forecasting, and radio resource provisioning.
- Safety-critical scenarios demand ultra-low latency and reliable communication.
- Current interfaces are fragmented across operators.
- Developers must understand complex radio layer and network topology (cells, frequencies, PLMNs), which slows innovation.

Goals:

- Enhance road safety by ensuring reliable communication between vehicles, infrastructure, and edge systems.
- Simplify APIs for non-telco developers, hiding radio/network complexity behind clean exposure interfaces.
- Improve emergency response routing with prioritized sessions (QoS/QoD) and real-time updates.
- Promote cross-operator interoperability through common, harmonized API semantics.

EXAMPLE OF EDGE RELEVANCE FOR SMART CITIES

The "**connected ambulance**" uses real-time connectivity and edge computing to optimize emergency routing with collision alerts, enable remote patient triage through streaming vitals and video to hospital physicians, and securely upload encrypted biometric and incident data to Electronic Health Records—ensuring faster response, early treatment, and seamless care continuity with full privacy compliance.



EXAMPLE: CONNECTED AMBULANCE SCENARIO

Collision Risk Detection and Safe Routing for Emergency Vehicles – real-time traffic data will be used to predict the drivers' behaviour, and critical alerts will be sent to nearby vehicles and VRUs, improving safety and speed of intervention in dense areas or during mass events

Relevant for
MEC V2X API

Real-Time Remote Triage and Diagnostics – a connected ambulance equipped with high-resolution cameras, medical sensors, and edge computing capabilities can stream patient vitals, ECGs, and high-definition video to emergency physicians at the hospital in real time

Relevant for
CAMARA QOD
API

Secure Biometric and Incident Data Upload to Electronic Health Records (EHR) – during patient transport, biometric and contextual data (e.g., accident scene video, environmental conditions) can be processed and temporarily stored at the edge for immediate decision support. Once securely validated, relevant data is selectively uploaded via trusted APIs to the patient's EHR, using privacy-aware access controls and real-time encryption mechanisms. This ensures continuity of care while respecting privacy and regulatory compliance.

(optionally: New
Secure DATA
Transfer API ?)

POSSIBLE EXAMPLE: MEC-CAMARA V2X API

To support the above use case, V2X related capabilities must be exposed both at MEC and CAMARA side

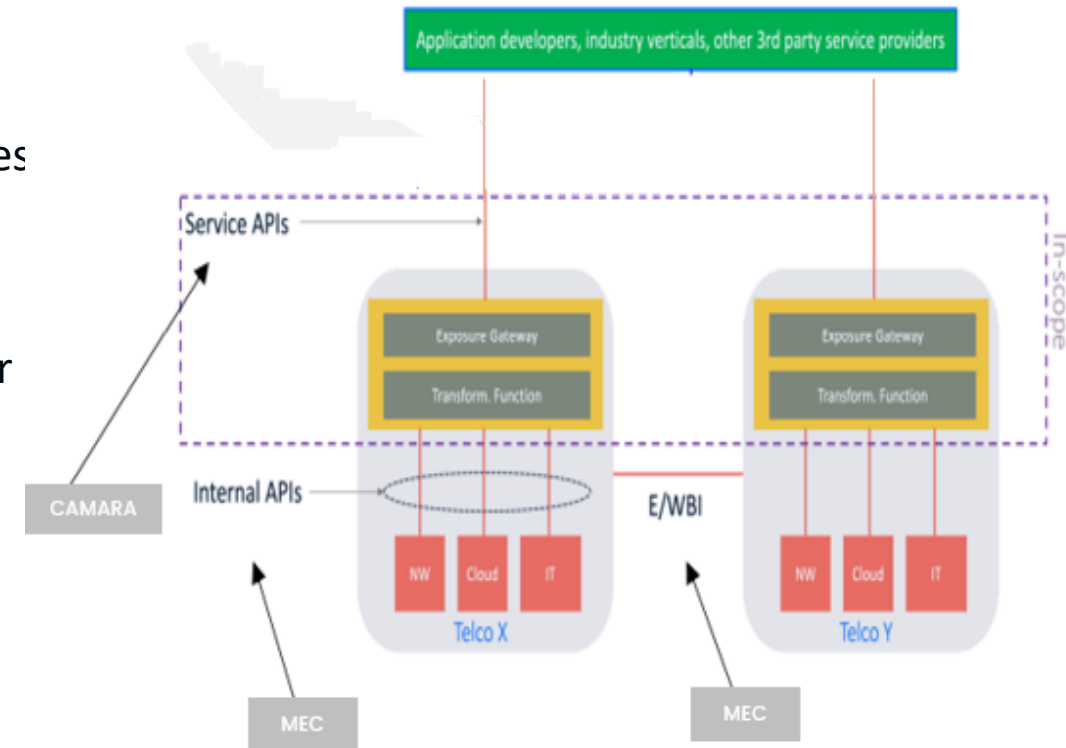
- ETSI MEC already defines a V2X Information Service API that enables communication between vehicles, roadside units, and MEC applications.
- This API support localized interactions such as:
 - Vehicle registration at the nearest edge node
 - Exchange of real-time traffic, speed, and safety messages
 - Hazard and congestion notifications to nearby vehicles

However, MEC APIs are edge-native and network-oriented, meaning they are not directly usable by application developers who expect simpler, service-level abstractions.

CAMARA, on the other hand, currently does not have any V2X-specific API. It focuses on generic capabilities like location, QoS, device status, connectivity prediction, etc.

PROPOSAL AT CAMARA SIDE

- Propose a new CAMARA V2X Exposure API, leveraging ETSI MEC V2X Information Service. Aligning both enables a unified, simplified experience across operators
- Implement a Transformation Function (TF) to align API attributes and methods between MEC SBI and CAMARA NBI.
- **Purpose:**
 - Expose standardized, operator-agnostic V2X capabilities for vehicles, RSUs, and edge applications.
 - Enable priority session establishment through CAMARA QoD/QoS APIs for emergency scenarios.
 - Facilitate priority passage and optimal route selection in emergency response cases.
 - Promote cross-operator interoperability by harmonizing MEC and CAMARA API semantics.



CONCEPTUAL API MAPPING – MEC o3o AND CAMARA

MEC API	CAMARA API (Proposed)	Description
/queries/uu_unicast_provisioning_info	/v2x/v1/provisioning	Provide a single-entry point for a vehicle to obtain all necessary network configurations for a specific geographic area.
/queries/uu_mbms_provisioning_info		
/queries/pc5_provisioning_info		
/provide_v2x_msg_distribution_server_info	/v2x/v1/brokers	Provide V2X Message Distribution Servers information required for direct connection with a service consumer.
/publish_v2x_message	/v2x/v1/messages	Publish a V2X message to VIS.
/vis/v2/subscriptions/*	/v2x/v1/subscriptions	A single entry point to subscribe for notifications on Uu Unicast, MBMS, PC5 provisioning changes, and V2X message updates.

PROVISIONING RESOURCE

- **CAMARA Endpoint:** GET /v2x/v1/provisioning
- **CAMARA Intent:**
 - Expose MEC-derived V2X provisioning information to application developers in a standardized, operator-agnostic way.
 - Provide a single unified API for a vehicle to obtain all required network configuration information for a specific geographic area.
 - Abstract away complex network details such as cell IDs, PLMNs, EARFCNs, frequencies, and radio technologies, enabling a simpler developer experience.
- **Description:**
 - The application sends a request with location parameters (e.g., latitude, longitude, radius) to CAMARA.
 - CAMARA invokes the Transformation Function (TF), which gathers relevant V2X provisioning data from MEC (e.g., unicast, multicast/MBMS, sidelink PC5).
 - The TF consolidates MEC-provided information and produces a harmonized, developer-friendly representation.
 - The developer receives a unified provisioning package without needing to understand underlying MEC resource paths or radio-layer complexity.

SCHEMA MAPPING BETWEEN MEC – CAMARA (PROVISIONING RESOURCE)

MEC V2X Representation

```
{
  // MEC UU Unicast provisioning
  // GET /queries/uu_unicast_provisioning_info
  "uuUnicast": [{
    "locationInfo": {
      "geoArea": { "latitude": 48.8584, "longitude": 2.2945 },
      "neighbourCellInfo": [// detailed neighbor info (FDD/TDD, PCI, PLMN, ECGI)],
      "v2xApplicationServer": {
        "ipAddress": "10.100.10.55",
        "udpPort": 5001}],
    // MEC UU MBMS provisioning
    // GET /queries/uu_mbms_provisioning_info
    "uuMbms": [{
      "locationInfo": {
        "geoArea": { "latitude": 48.8584, "longitude": 2.2945 },
        "neighbourCellInfo": [// detailed neighbor info with MBMS service area identity],
        "v2xServerUsd": {
          "sdpInfo": {
            "ipMulticastAddress": "239.192.10.1",
            "portNumber": 5000 },
          "tmgi": { "mcc": "001", "mnc": "01", "mbmsServiceId": "0xA1B2C3" } }},
    // MEC PCS provisioning
    // GET /queries/pc5_provisioning_info
    "pc5": [{
      "locationInfo": {
        "geoArea": { "latitude": 48.8584, "longitude": 2.2945 },
        "neighbourCellInfo": [ // neighbor info for sidelink (optional)],
        "dstLayer2Id": "....."}}]}
```

Proposed CAMARA Representation

```
// CAMARA
{
  "requestedLocation": { "lat": 48.8584, "long": 2.2945 },
  "timestamp": "2025-12-05T17:00:00Z", // from MEC timeStamp (ISO 8601)

  "unicast": {
    "serverAddress": "10.100.10.55", // from MEC v2xApplicationServer.ipAddress
    "serverPort": 5001, // from MEC v2xApplicationServer.udpPort
    "protocol": "udp"
  },

  "multicast": {
    "tmgi": "0xA1B2C3", // from MEC tmgi.mbmsServiceId (+MCC/MNC)
    "multicastAddress": "239.192.10.1", // from MEC sdpInfo.ipMulticastAddress
    "port": 5000 // from MEC sdpInfo.portNumber
  },

  "sidelink": {
    "destinationLayer2Id": "....." // direct from MEC dstLayer2Id
  }
}
```

PROVISIONING RESOURCE FLOW WITH TRANSFORMATION FUNCTION

- As an example, we can take V2X API from MEC and V2X API from CAMARA.
- App → CAMARA: GET /v2x/v1/provisioning
- CAMARA → Transformation Function (TF): normalize attributes & methods
- TF → MEC: GET /queries/*
 - TF maps GET → GET, aligns attributes, ensures interoperability
- Note: CAMARA GET /v2x/v1/provisioning can invoke one or multiple MEC endpoints (uu_unicast, uu_mbms, pc5) depending on the query

CONCLUSION

Why join this initiative?

- **A shared opportunity for impact and collaboration**
- **International visibility & recognition** through CAMARA and ETSI-aligned standardization activities
- **Showcasing real smart city deployments**, strengthening the positioning of Turin and project partners
- **Stronger ecosystem collaboration**: moving from isolated initiatives to joint, interoperable solutions
- **Future joint opportunities** enabled by cross-project synergies (e.g., ToMOVE, ENVELOPE)
- **Technical advancement** through knowledge sharing and real-world API evolution

What we ask from you?

- **Support dissemination and advocacy** at institutional, policy, and communication levels
- **Facilitate alignment with ongoing initiatives** and smart city strategies
- **Provide technical feedback and peer review** to strengthen the proposal before submission
- Contribute to shaping a **shared and interoperable API framework**



THANKS!

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